

FC 1: Elex

Number	Question	Answer
1	What is the input impedance of the second stage when cascading amplifiers?	The input impedance of the second stage is the load for the first stage. For a high gain, it is generally designed to be high.
2	Which theorem relates to the behaviour of linear, passive, and bilateral circuits and states that the ratio of voltage to current remains unchanged when the positions of the source and response are interchanged?	Reciprocity Theorem
3	In a power triangle, what is the relationship between active power (P), reactive power (Q), and apparent power (S)?	$S^2 = P^2 + Q^2$
4	What law states that a time-varying magnetic field creates a circulating electric field?	Faraday's Law of Electromagnetic Induction
5	How can you improve the power factor of an AC inductive circuit?	Add a parallel capacitor.
6	Coulomb's Law states that the electrostatic force between two point charges is directly proportional to what?	The product of the magnitudes of the charges.
7	In a common-source FET amplifier, what is the effect of increasing the source degenerating resistor?	It decreases the voltage gain and increases the linearity and stability.
8	What does Maxwell's first equation (Gauss's Law for electricity) state?	The electric flux through a closed surface is proportional to the total electric charge enclosed within that surface.
9	What is the induced EMF in a conductor placed in a uniform, static magnetic field?	Zero
10	For a cylindrical conductor with radius r placed with its axis perpendicular to a uniform electric field, what is the magnitude of the electric field at the conductor's surface?	The electric field is zero
11	In a center-tapped full-wave rectifier, what happens to the DC output voltage under load?	It becomes lower than the peak secondary voltage due to diode drops and transformer regulation.
12	How does non-uniform current density (J) vary with radius r in a cylindrical conductor?	proportional to the radius
13	In a half-wave rectifier, the DC output is nearly zero. What is the most probable cause?	The diode is open-circuited.

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14	In a voltage doubler circuit, the output voltage is lower than expected. What is the most probable cause?	The capacitors discharge too quickly under load (insufficient capacitance or too low load resistance).
15	What is the advantage of a Schottky Diode in high-frequency applications?	It has a very fast switching speed and lower forward voltage drop, which reduces switching losses.
16	In a full-wave rectifier with a resistive load and filter capacitor, the output DC voltage is significantly lower than expected. What is the most probable cause?	The filter capacitor has too small a value, leading to excessive ripple.
17	In a Zener diode voltage regulator, the input voltage drops significantly with a load current of 30mA. What is the most probable cause?	The load current has exceeded the maximum current the Zener can regulate, or the series resistor is too large.
18	In a precision rectifier circuit, the output voltage is lower than expected. What is a likely cause?	The op-amp's open-loop gain is insufficient or the circuit is designed incorrectly.
19	In a precision rectifier circuit, the output signal is lower than expected and shows distortion. What is the most likely cause?	The op-amp has an insufficient slew rate.
20	Which component smoothens the pulsating DC after rectification?	Filter Capacitor
21	When does an N-type enhancement MOSFET start to conduct?	When the gate-source voltage (V_{GS}) exceeds the threshold voltage (V_{TH}).
22	In an N-channel MOSFET, the gate-source voltage is below the threshold. What is the operating region?	Cut-off region.
23	In a BJT, if both the base-emitter (BE) and collector-base (CB) junctions are reverse-biased, what is the operating region?	Cut-off region.
24	What component in a common-emitter amplifier provides negative feedback to help stabilize the DC gain?	The emitter resistor.
25	What is the operating region of a BJT when both the emitter-base and collector-base junctions are reverse-biased?	Cut-off region.

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26	How can high-frequency roll-off in a MOSFET amplifier caused by the Miller effect be reduced?	By minimizing the gain of the amplifier stage that contributes to the Miller capacitance.
27	What controls the drain current in a MOSFET when it is operating in the ohmic (triode) region?	The gate-source voltage (V_{GS}).
28	What happens when the drain-source voltage (V_{DS}) is increased beyond the ohmic region in a MOSFET?	The MOSFET enters the saturation (active) region.
29	What happens when a MOSFET operates in the cut-off region?	The drain current is approximately zero.
30	Why is the gate-source voltage (V_{GS}) of an amplifier sometimes used to control a Voltage-Controlled Oscillator (VCO)?	To vary the VCO's output frequency, as V_{GS} can control a variable capacitance or current.
31	What happens to a MOSFET's Threshold Voltage (V_{TH}) as temperature increases?	It decreases.
32	In a common-emitter amplifier, what is the primary effect of adding a bypass capacitor across the emitter resistor?	It increases the AC voltage gain by bypassing the AC signal to ground.
33	What is a common symptom when the supply voltage (V_{CC}) is too low in a circuit?	Circuit malfunction, reduced output swing, or no operation.
34	What is the small-signal resistance of a forward-biased diode called?	Dynamic Resistance (or AC resistance).
35	In a diode, which capacitance is formed by the depletion region acting as the dielectric between the p and n regions?	Junction Capacitance (or Depletion Capacitance).
36	An engineer wants to find the small-signal resistance of a diode operating at 2mA. What are they calculating?	The dynamic resistance.
37	In a diode, which capacitance increases when the reverse bias voltage is increased?	Junction Capacitance (Depletion Capacitance decreases with reverse bias; the statement is incorrect. Diffusion capacitance is significant in forward bias).
38	In a diode, what represents the resistance present in the semiconductor bulk and contacts?	Series Resistance.
39	What is the function of the ideal diode in a piecewise linear model?	It represents perfect conduction in one direction and perfect blocking in the other.

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40	What is the primary use of a voltage multiplier circuit?	To step up an AC voltage to a higher DC voltage without using a transformer.
41	What type of circuit, composed of diodes and capacitors, converts a low AC voltage to a higher DC voltage?	Voltage Multiplier.
42	What circuit converts AC to DC?	Rectifier.
43	In a BJT's normal active operation, how is the base-emitter junction biased?	Forward-biased.
44	What component in a power supply maintains a constant DC output voltage despite variations in input or load?	Voltage Regulator.
45	In a linear DC power supply, what component steps up or down the AC input voltage?	Transformer.
46	In a MOSFET, if the drain-source voltage is small, in what region does it operate?	Ohmic (or Triode) Region.
47	In a linear power supply, what is the purpose of the voltage regulator?	To maintain a constant DC output voltage.
48	Why is a Zener diode mostly used as a voltage regulator?	It maintains a nearly constant voltage across it when operated in its reverse breakdown region.
49	A UJT is heavily doped. What is the most probable reason for this in a triggering circuit for an SCR?	To achieve a stable and consistent peak point voltage.
50	What is the main advantage of a Switched-Mode Power Supply (SMPS) over a linear supply?	Higher efficiency and lower heat generation.
51	Light travels from air ($n=1.0$) to water ($n=1.33$). How does it bend?	Towards the normal.
52	What components are used in an SMPS to transfer and store energy?	Transformer and Inductor.
53	What is the goal of Resource Allocation?	To maximize the use of available resources.
54	What happens to the gain of a common-emitter amplifier if the emitter resistor is removed?	The gain increases, but DC stability decreases.
55	In a BJT common-emitter amplifier, if the emitter bypass capacitor is removed, what happens?	The voltage gain decreases due to increased negative feedback.
56	What is the SI unit for Angular Velocity?	Radian per second (rad/s).
57	Removing the bypass capacitor in the emitter of a common-emitter amplifier introduces what, which lowers the voltage gain?	Negative feedback.

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58	What is a cost that has already been incurred and cannot be recovered?	Sunk Cost.
59	What is a key characteristic of a common-gate FET amplifier?	Low input impedance.
60	In a MOSFET, what is the primary control variable for the drain current?	Gate-source voltage (V_{GS}).
61	In a UJT, when the emitter current increases after the peak point, what happens to the emitter resistance?	It decreases.
62	In a diode model, what component represents the resistance in the semiconductor material and contacts?	Series Resistor.
63	In a piecewise linear diode model, what does the "ideal" component represent?	Perfect, instantaneous switching with zero forward voltage and infinite reverse resistance.
64	What type of circuit can step up voltage without a transformer?	Voltage Multiplier.
65	In a voltage multiplier circuit, why is the output voltage under load less than the theoretical value?	Due to diode forward voltage drops, capacitor leakage, and finite capacitance.
66	What is a common application of a UJT?	Relaxation oscillators and pulse generators.
67	What is the peak point voltage of a UJT?	The emitter voltage at which the UJT begins to conduct heavily and enter the negative resistance region.
68	What solid-state device acts as a bidirectional trigger for another semiconductor switch like a TRIAC?	DIAC.
69	What determines the response time of a photodiode?	The RC time constant, primarily influenced by the junction capacitance and load resistance.
70	What is the peak point voltage in a UJT?	The voltage at which the emitter current starts to rise sharply and the device enters the negative resistance region.
71	What is a common application of a UJT?	Relaxation oscillator and pulse generator.

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72	Two op-amp configurations have gains of 1 and 10. Which has a lesser bandwidth?	The configuration with a gain of 10 (higher gain results in lower bandwidth).
73	What determines the closed-loop gain of an operational amplifier circuit?	The external feedback resistor network.
74	What is the main function of a UJT?	To act as a triggering device in relaxation oscillators.
75	In what region does a UJT operate after the peak point?	Negative Resistance region.
76	What happens to the AC voltage gain of an amplifier when the load resistance increases?	It increases.
77	What is the effect of parasitic capacitance in FET circuits?	It reduces the bandwidth.
78	What is the effect of load impedance on the voltage gain of an amplifier?	Higher load impedance results in higher voltage gain.
79	Why does source impedance affect the voltage gain of an amplifier?	It forms a voltage divider with the amplifier's input impedance, reducing the actual voltage at the input terminals.
80	What is a Chebyshev filter mainly characterized by?	Ripples in the passband and a steeper roll-off compared to a Butterworth filter.
81	In a differential amplifier, what is the output if the same signal is applied to both inputs?	The ideal output is zero (common-mode rejection).
82	In a constant current limiter circuit, as the output voltage increases, what happens to the output current?	It remains constant (by design of the current limiter).
83	What is a characteristic of the output response of a Chebyshev filter?	It has ripples in the passband and a sharp roll-off.
84	What is a key characteristic of a Bessel filter?	It has a linear phase response, minimizing waveform distortion.
85	For a voltage amplifier, what condition between output resistance (R_O) and load resistance (R_L) maximizes the voltage across the load?	$R_O \ll R_L$.
86	What characterizes a Chebyshev filter?	Ripples in the passband, which allow for a steeper roll-off than a Butterworth filter.
87	What causes a decrease in gain at high frequencies in a BJT amplifier?	The internal junction capacitances.

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88	In a current mirror, if the reference current increases, what happens to the output current?	It increases proportionally.
89	In a differential amplifier, if the same voltage is applied to both inputs, what is the output?	Zero (ideal case) or a very small voltage (due to finite CMRR).
90	What is the input impedance of a cascade amplifier primarily determined by?	The input impedance of the first stage.
91	What is the force per unit length between two long parallel wires carrying currents I_1 and I_2 , separated by distance d ?	$(\mu_0 * I_1 * I_2) / (2\pi d)$
92	What is the primary purpose of a DIAC?	To act as a bidirectional trigger device for TRIACs in AC power control circuits.
93	What is the SI unit of magnetic flux?	Weber (Wb). (Tesla is the unit of flux density).
94	What is the primary purpose of the drain current in a MOSFET?	To act as the output current that is controlled by the gate-source voltage.
95	What is the formula for the coupling coefficient (k) in terms of mutual inductance (M) and self-inductances (L_1, L_2)?	$k = M / \sqrt{L_1 * L_2}$
96	What is the use of a compensation capacitor in an op-amp?	to act as the output current of the MOSFET, which is controlled by the voltage applied to the gate, V_{GS}
97	What is the Common Mode Rejection Ratio (CMRR)?	The ratio of the differential gain to the common-mode gain.
98	The CMRR of a differential amplifier is the ratio of what?	The differential gain (A_d) to the common-mode gain (A_{cm}).
99	What is the main purpose of a compensation capacitor in an op-amp?	To ensure stability by dominating the frequency response and providing a controlled roll-off.
100	What primarily determines the frequency response of an op-amp?	The internal compensation and the gain-bandwidth product (GBW).
101	What is the primary purpose of an optocoupler?	To provide electrical isolation between the input and output circuits.
102	What type of filter is used to remove a single frequency or a narrow band of frequencies?	Notch Filter (or Band-Stop Filter).

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103	What is the primary purpose of a compensation capacitor?	To ensure stability in feedback circuits.
104	For two amplifiers in cascade, what primarily determines the overall input impedance?	The input impedance of the first stage.
105	What is the Butterworth Filter primarily characterized by?	A maximally flat passband response (no ripple).
106	What is the main advantage of a Bessel Filter?	Linear phase response, which preserves the shape of signals in the passband.
107	What parameter does a compensation capacitor primarily affect?	Phase Margin (and thus stability).
108	What determines the frequency response of an op-amp?	Its open-loop gain and the gain-bandwidth product.
109	For a second-order band-pass filter, how is the quality factor (Q) defined?	$Q = f_{\text{center}} / \text{Bandwidth}$
110	In an RC phase-shift oscillator, what is the total phase shift required around the feedback loop?	360 degrees (or 0 degrees).
111	Which oscillator can be used to have sustained oscillations?	Hartley Oscillator (among others like Colpitts, Phase-Shift, etc.).
112	The frequency response of a bandpass filter is _____ within the frequency band and _____ outside it.	Maximum; Attenuated (or Zero)
113	What mainly affects the high-frequency response of a BJT amplifier?	The internal junction capacitances.
114	In a second-order active band-pass filter, if Q is increased while f_0 is constant, what happens?	The bandwidth narrows and the amplitude at f_0 increases.
115	The maximum frequency response of a bandpass filter is _____ within it and _____ outside it.	Maximum; Minimum (or Zero)
116	What primarily affects the bandwidth of a common-emitter BJT amplifier?	The internal capacitances and the external resistive loads.
117	What is a UJT commonly used for?	As a triggering device for SCRs and TRIACs.
118	What is the difference between a Current Source and a Current Mirror?	A current source provides a constant current, while a current mirror replicates (mirrors) a reference current.
119	What compensation technique is added to op-amp circuits to control frequency response?	Miller Compensation.

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120	What is a key difference between a current mirror and a simple current source?	A current mirror copies a reference current, often with scaling, while a current source generates a fixed current.
121	In an amplifier, what is the pole frequency?	The frequency at which the gain decreases by 3 dB from its maximum value.
122	How do you calculate the center frequency of a bandpass filter?	Compute the average of lower and upper cutoff frequencies
123	Which law states that the magnetic field circulation around a closed loop is equal to μ_0 times the enclosed current?	Ampère's Circuital Law.
124	How does a snubber circuit protect a thyristor?	By limiting the rate of rise of voltage (dv/dt) across it.
125	In which region does a BJT operate as a current-controlled current source?	Active Region.
126	What is a disadvantage of Nickel-Cadmium (Ni-Cd) batteries?	Memory Effect.
127	What circuit converts a low AC voltage to a higher DC voltage using diodes and capacitors?	Voltage Multiplier.
128	What is the total inductance of two coils connected in series aiding?	$L_{\text{total}} = L_1 + L_2 + 2M$
129	In a BJT's normal active operation, how is the emitter-base junction biased?	Forward-biased.
130	What is the total inductance of two coils connected in series opposing?	$L_{\text{total}} = L_1 + L_2 - 2M$
131	What device is equivalent to two SCRs connected in inverse parallel?	TRIAC.
132	What is the system response when the damping factor is greater than 1?	Overdamped.
133	What does a Standing Wave Ratio (SWR) measure in a transmission line?	The impedance match between the transmission line and the load.
134	How many RC networks are needed in a phase-shift oscillator to achieve a 180° phase shift?	Three.
135	In a TRIAC-based lamp dimmer, what happens if the firing angle is increased?	The lamp's brightness decreases.
136	All are examples of optoelectronics except: a) piezoelectric, b) optocoupler, c) LED, d) photodiode.	a) Piezoelectric.

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137	How does a Light Emitting Diode (LED) produce light?	Through photon emission from electron-hole recombination in a forward-biased semiconductor.
138	What is the main advantage of a Gate Turn-Off Thyristor (GTO) over a conventional SCR?	It can be turned off by a negative gate signal.
139	What is a common application of a GTO?	High-power AC and DC motor drives.
140	What is the phase angle in a series RLC circuit?	$\theta = \arctan((X_L - X_C) / R)$
141	When does a DC source deliver maximum power to the load?	When the load resistance equals the source's internal resistance.
142	Which theorem states that the response in a linear circuit with multiple sources is the sum of responses from each source acting alone?	Superposition Theorem.
143	What is the unit of electric field strength?	Volts per meter (V/m). (Coulomb/m ² is the unit for electric flux density).
144	What is the formula for the coupling coefficient (k)?	$k = M / \sqrt{L1 * L2}$
145	If the leakage flux in a transformer's secondary winding increases, what happens to the output voltage?	The output voltage decreases.
146	When the air gap length of a magnetic circuit is doubled while the MMF is constant, what happens to the magnetic flux?	The flux becomes roughly half its original value.
147	Magnetic reluctance ($\mathfrak{R}$) is inversely proportional to what material property?	Permeability (μ).
148	In a two-port network's Y-parameters, what does y_{12} represent?	The reverse transfer admittance (I_1 / V_2 with $V_1=0$).
149	What is the total inductance of two coils connected in series aiding?	$L1 + L2 + 2M$
150	Why might two coils have zero mutual inductance ($M=0$)?	Because their magnetic fields are orthogonal (not linked) or they are far apart.
151	When analyzing the resistance of a non-uniform conductor, what is the first step?	To determine how the cross-sectional area varies along its length.
152	How does a small air gap affect the total reluctance of a high-permeability magnetic core?	It drastically increases the total reluctance.
153	What type of filter has a maximally flat frequency response in the passband?	Butterworth Filter.
154	What type of filter has a linear phase response?	Bessel Filter.

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155	As temperature increases, what happens to the electrical resistance of a semiconductor?	It decreases.
156	In a current mirror, if the reference current increases, what happens to the output current?	It increases proportionally.
157	What are the basic components of an SCR snubber circuit?	A resistor and a capacitor in series.
158	According to Thevenin's Theorem, any linear bilateral network can be replaced by an equivalent circuit composed of what?	A single voltage source in series with a resistor.
159	What is the electric field outside a uniformly charged sphere ($r > R$)?	$E = Q / (4\pi\epsilon_0 r^2)$
160	What is the operating region of a BJT with the base-emitter junction forward-biased and the collector-base junction reverse-biased?	Active Region.
161	What is the operating region of a BJT with both the base-emitter and collector-base junctions forward-biased?	Saturation Region.
162	What is a bidirectional, two-terminal thyristor that can be triggered into conduction in both directions?	TRIAC.
163	What is a bidirectional, two-terminal diode-like trigger device?	DIAC.
164	What is the primary use of a voltage multiplier?	To generate a high DC voltage from a lower AC voltage without a transformer.
165	In a power supply, what component is used to smooth the pulsating DC from the rectifier?	Filter Capacitor.